

MTA Ridership Patterns

Power BI Specialist Final Project | Digital Egypt Pioneers Initiative (DEPI)

11/29/25

Meet Our Project Team



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Agenda

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Rider Trends and Recovery

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Problem Context: COVID-19 and Public Transport

The **COVID-19** pandemic disrupted public transport systems worldwide, with a significant impact on MTA in New York.

- Sharp decrease in ridership due to lockdowns and remote work
- Financial pressures and uncertainty for transport agencies

Need to understand **new travel behaviors** instead of assuming a full return to old patterns

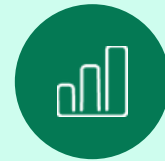


Project Goals



Main Goal

Analyze MTA rider patterns **before and after COVID-19** using Power BI, and understand the "new normal"



Time Comparison

Compare daily rider levels **pre-COVID (2019)** with **post-COVID**



Measure Recovery

Measure **rider recovery** by transport type (subway, buses, bridges and tunnels)



Daily Patterns

Identify **weekday vs. weekend** patterns, including the **Friday effect**



Recommendations

Provide **data-driven recommendations** for fares, services, and marketing

Business Impact and Motivation



7.90M **4.65M** **41%**

Average Daily Ridership Pre-COVID **Average Daily Ridership Post-COVID** **Ridership Loss**

2019 Baseline Level Current Transit Situation Significant Decrease in Usage



Estimated Financial Impact: \$21.81 Billion

These changes directly affect MTA's ability to fund operations, maintenance, and future investments.

Data Structure and Key Fields

Key Columns

Field	Description
Date	Daily Date
Mode / System	Mode of Transport
Daily Ridership	Daily Ridership
2019 Baseline	2019 Baseline
Recovery %	Recovery %

Included Transportation Modes

- Subways
- Buses
- LIRR
- Metro-North
- Bridges & Tunnels
- Access-A-Ride

The data is a daily **time series**, allowing for detailed trend analysis.

Data Cleaning and Preparation



Import

Import raw CSV files into Power BI



Standardize

Standardize column names and data types (dates, numbers,



Process

Process missing or inconsistent values



Filter

Filter out invalid or out-of-scope records

"From raw CSV to an analysis-ready model"

This ensures **reliable performance indicators and accurate visualizations** for comprehensive analysis.

Modeling and Calculations (DAX)

Key DAX Metrics

Total Ridership

Total number of passengers

Average Daily Ridership

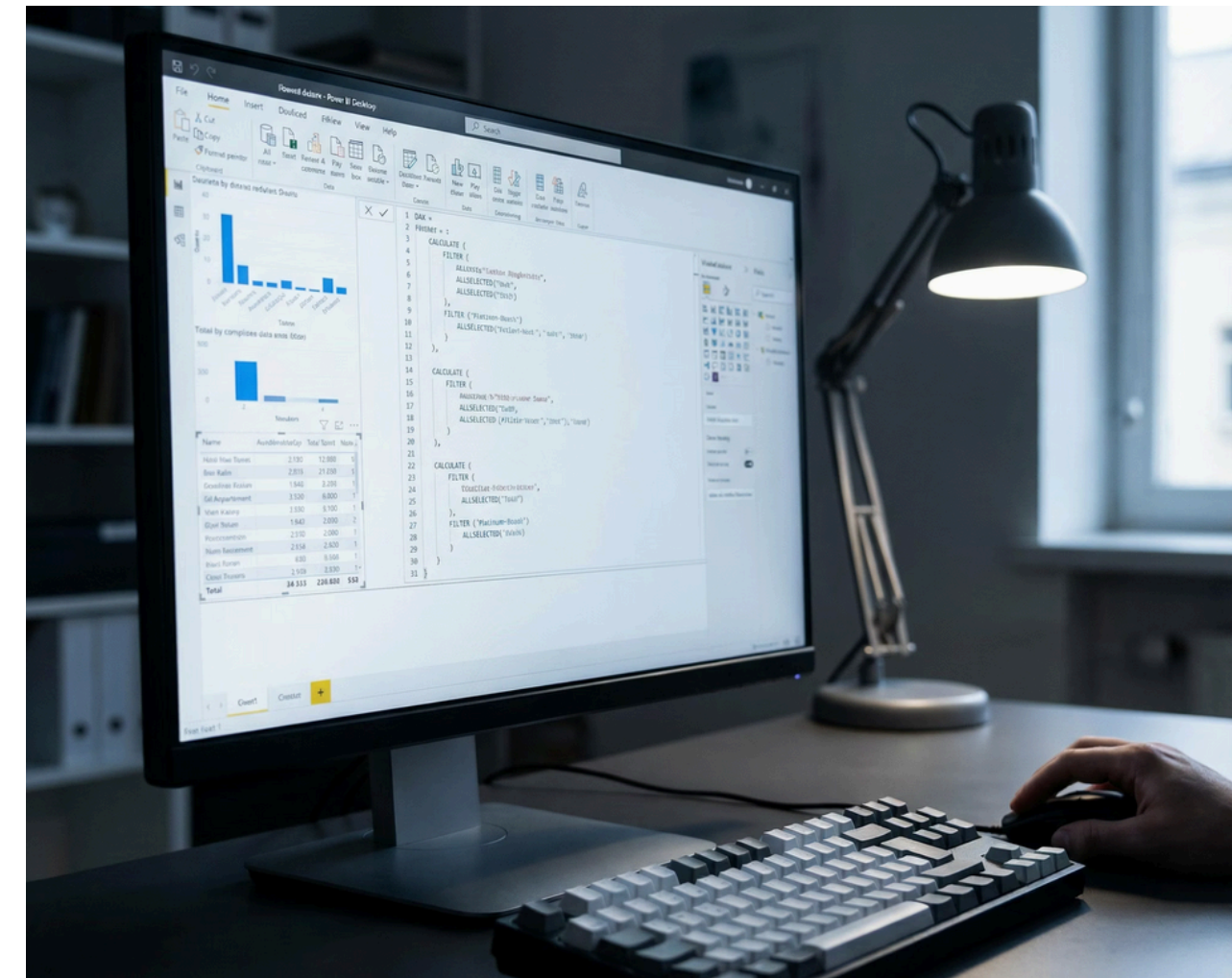
Average daily passengers

Recovery % vs 2019

Recovery percentage compared to 2019

Ridership Difference

Absolute and relative difference in passengers



Overall Ridership Trends

Key Messages

Significant drop from **7.90 million** to **4.65 million** daily riders

- Recovery is partial and uneven over time
- Ridership has not fully returned to 2019 levels

"Ridership has not fully recovered to 2019 levels, suggesting a structural change rather than a temporary shock."

Mode Recovery: The Big Picture

54.17%

Subway

Moderate recovery for public transit

93.36%

Bridges & Tunnels

Near-complete recovery for cars

Key Insight

Public transport (subway, buses) is slow to recover, while car-related infrastructure shows near-complete recovery.



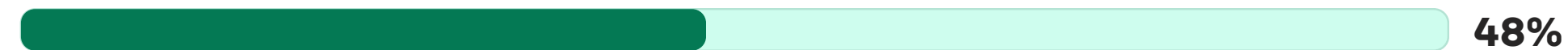
Recovery of Modes: Focus on Public Transit

Detailed Comparison



Subways

Subways



Buses

Buses



LIRR

Long Island Rail Road



Metro-North

Metro-North



What Does This Mean?

- Which public transit mode is recovering faster?
- Which was most affected by the pandemic?

Impact on [service planning](#) and operations



Modal Recovery: Bridges and Tunnels / Car Behavior

93.36%

Bridges and Tunnels Recovery

Strong Shift Towards Cars

High recovery (~93%+) indicates a strong preference for car use after COVID.

Possible Reasons

Perceived safety, comfort, flexibility in personal travel.

Impacts

Traffic congestion, environmental impact, need for incentives to return to public transport.

Weekday Patterns Overview

1

Monday

Lower than expected

2

Tuesday - Thursday

Highest - Peak of the week

3

Friday

Significant drop

4

Weekend

Different patterns



❏ "Traditional 5-day commute shifted to a 3-day in-office core (Tuesday-Thursday)."

In-Depth Analysis of the "Friday Effect"

What is the Friday Effect?

Many workers choose to work remotely or have flexible schedules on Fridays, leading to:

- Reduced demand at the end of the work week
- A shift in traditional commuting patterns

❑ **The "Friday Effect":** Ridership consistently peaks Tuesday to Thursday, then falls dramatically on Friday. This visible, repeatable pattern proves hybrid work is a permanent fixture.

Weekend Behavior and Recovery

66.73%

Sunday

Highest recovery of the week

Explanation

Stronger relative recovery on weekends suggests **more leisure and non-work trips**.

55.25%

Monday

Lowest recovery of the week

Opportunities

Potential for **weekend-focused products** (e.g., day tickets, family offers).

Summary of Key Insights

Ridership Significantly Lower than 2019

Ridership remains **much lower than 2019 levels**, with major revenue impacts.

Public Transport Recovery is Slower

The recovery of **public transport** (metro, buses) is much slower than **car-related travel**.

Hybrid Work Created a New Pattern

Hybrid work created a noticeable **Tuesday-Thursday core** and a **Friday effect**.

Weekends are Stronger

Weekends, especially Sundays, show **relatively higher recovery**, linked to leisure and non-commute trips.

Recommendations



Pricing & Products

Offer flexible tickets for **3-day commute weeks**

- Introduce weekend passes / family bundles to capture leisure demand



Service & Operations

Align frequency with **peak days (Tuesday-Thursday)**

- Reduce excess capacity on low-demand days where appropriate
- Consider targeted improvements where recovery is stronger



Marketing & Communication

Promote public transit as **safe, reliable, and convenient** for both work and leisure

Run special campaigns around **weekends** and **Fridays**

- Highlight environmental and economic benefits

Thank You

Deliverables

- Comprehensive interactive Power BI report
 - In-depth data analysis of passenger patterns
 - Actionable insights and strategic recommendations
 - Dynamic visualizations and dashboards
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